WASP-52b

R-1904

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_190806` · created 2026-08-05T23:04:07+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458701.8098 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.209 +/- 0.061
Transit depth (Rp/Rs)^2	4.36 +/- 2.55 [%]
Orbital Inclination (inc)	87.6 +/- 2.77
Airmass coefficient 1 (a1)	0.93 +/- 0.58
Airmass coefficient 2 (a2)	0.01 +/- 0.49
Scatter in the residuals of the lightcurve fit is	64.07 %
Best Comparison Star	#2 - [418, 142]
Optimal Aperture	3.15
Optimal Annulus	6.57
Transit Duration (day)	0.084 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.209 ± 0.061 vs 0.1646 (deviation 0.69σ)
Duration fit vs published	0.084 d vs 0.0758 d (deviation 0.64σ)
Mid-time O–C	-11.7 min
Depth SNR	3.2
Residual scatter	64.07 %

Light curve

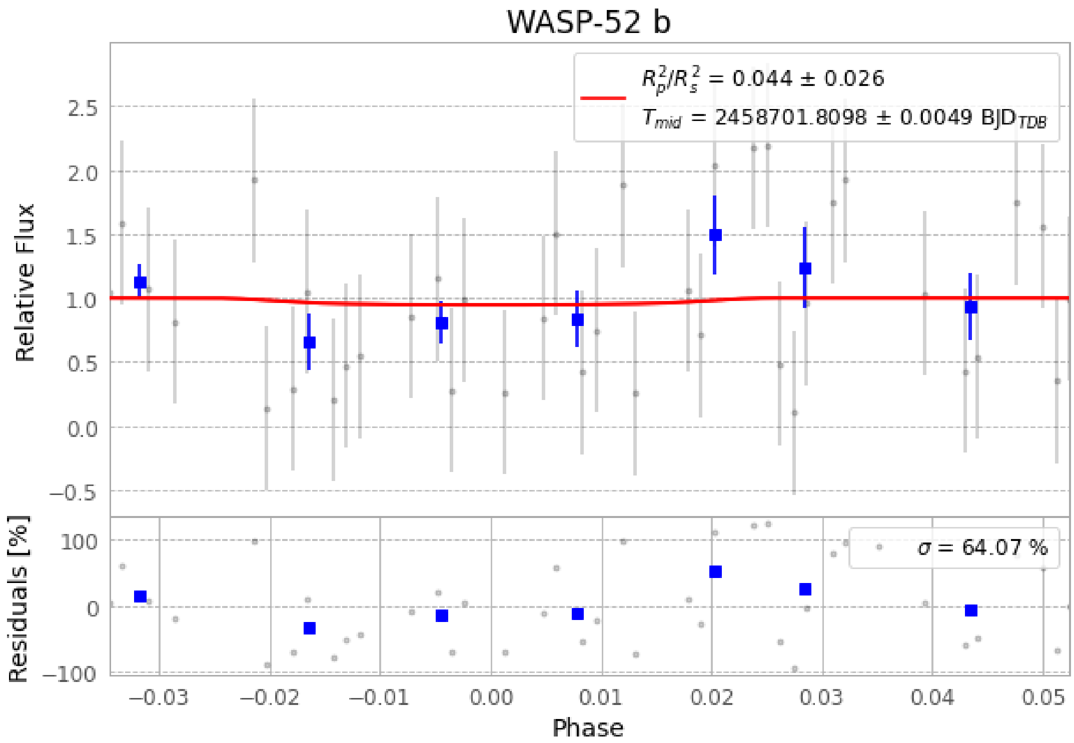

Figure 1 — FinalLightCurve_WASP-52 b_2019-08-05.png (SHA-256 09a5616ecfecc020…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [151, 313], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 d35c51cbdfe9e0fa…).
`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `WASP-52190806054212.FITS` → `WASP-52190806093012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-190806.log (f826bcb89a9a…), FinalLightCurve_WASP-52 b_2019-08-05.png (09a5616ecfec…), FinalLightCurve_WASP-52 b_2019-08-05.csv (336199a6a447…)

Compute resources

Wall time 136.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-05 23:04Z.